

FYS-STK3155/4155 September 21



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logistic Regression

linear regression

$$y_i = \underbrace{f(x_i)}_{\text{continuous function}} + \varepsilon_i \quad \begin{array}{l} \text{non-} \\ \text{stochastic} \end{array} \quad \varepsilon_i \sim N(0, \sigma^2)$$

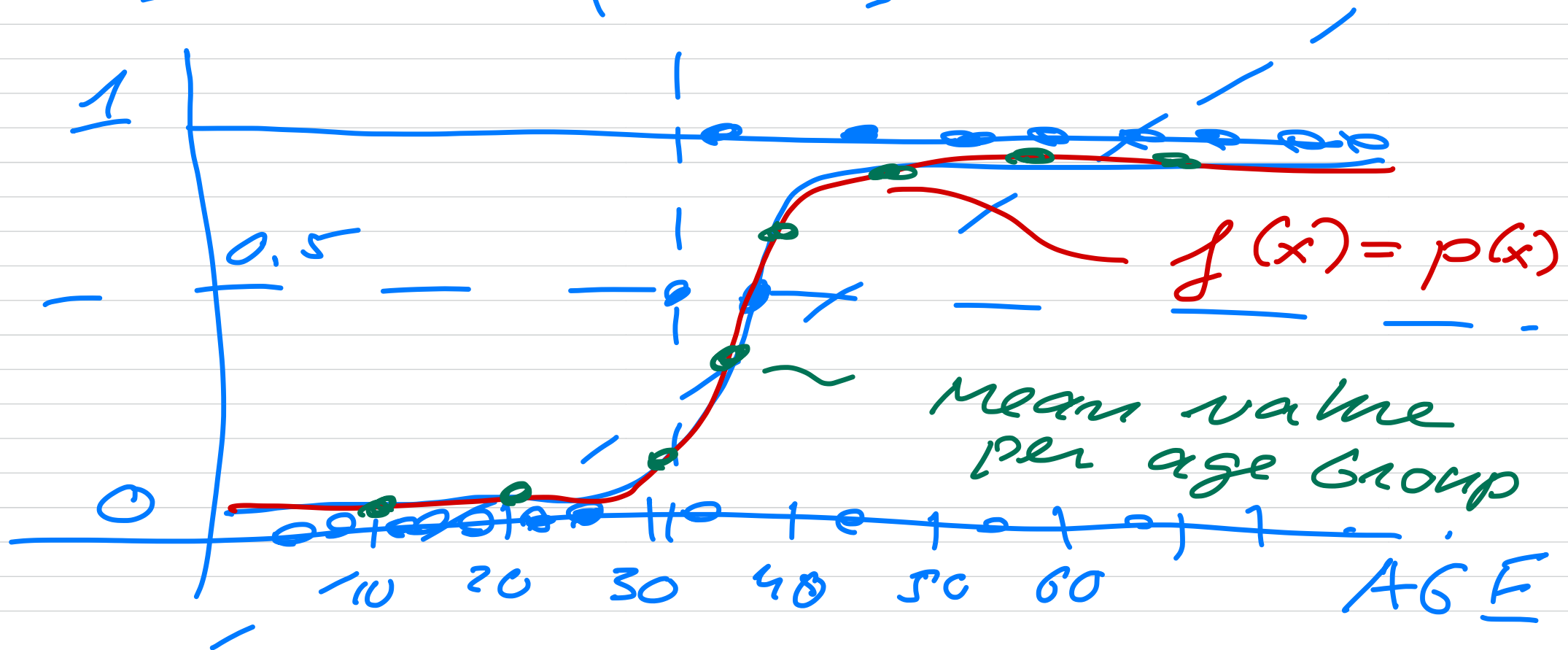
$\left(\begin{array}{l} \text{continuous} \\ \text{function} \end{array} \right) \rightarrow N(E[y_i], \sigma^2)$

$$f(x_i) \approx \sum_{j=0}^{p-1} x_{ij} \theta_j = X_{i*} \vec{\theta}$$
$$E[y_i] = X_{i*} \vec{\theta}$$

$$y_i \in (-\infty, \infty) \wedge x_i \in (-\infty, \infty)$$

what if (Binary)

$$y_i \in \{0, 1\}$$



$$y_i' = p(x_i') + \varepsilon_i'$$

$$p(x_i') = \frac{e^{\theta x_i'}}{1 + e^{\theta x_i'}}$$

linear in θ

$$p(x_i') = P_i' = \boxed{\frac{e^{\theta_0 + \theta_1 x_i'}}{1 + e^{\theta_0 + \theta_1 x_i'}}}$$

if $y_i = 1$, then $p(x_i') = P_i'$

$$\int_{x \in D} P_X(x) dx = 1$$

$$\left(\sum_{i \in D} P_X(x_i) = 1 \right)$$

$$y_i = 1 \quad \wedge \quad \frac{P_i}{P_1} = \frac{e^{G_0 + G_1 x_i'}}{1 + e^{G_0 + G_1 x_i'}}$$

$$y_i = 0 \quad \wedge \quad \frac{1 - P_i}{P_0} = \frac{1}{1 + e^{G_0 + G_1 x_i'}}$$

$$P_0 + P_1 = 1$$

which probability function (PDF) does ε_i follow?

$$y_i = p_i + \varepsilon_i$$

$$y_i = 1 \Rightarrow \varepsilon_i = 1 - p_i$$

$$y_i = 0 \Rightarrow \varepsilon_i = -p_i$$

$$(E[X] = \sum_{i=0}^n x_i P_X(x_i))$$

$$E[\varepsilon_i] = \underbrace{(1-p_i)}_{\varepsilon_i} \overset{\substack{\uparrow \\ y=1}}{p_i} + (-p_i) \underbrace{(1-p_i)}_{y=0} = 0$$

$$\text{var} [\epsilon_i] = (1-p_i)^2 p_i + (-p_i)^2 (1-p_i)$$

$$= p_i (1-p_i)$$

which follows a binomial distribution $\mathcal{D}$

$$\mathcal{D} = \{ (x_0, y_0), (x_1, y_1), \dots, (x_{n-1}, y_{n-1}) \}$$

$$P(\mathcal{D} | e) = \prod_{i=0}^{n-1} p_i^{y_i} (1-p_i)^{1-y_i}$$

$$y_i = \{0, 1\}$$

$$\hat{\theta} = \arg \min_{\theta \in \mathbb{R}^P} -\log P(\mathcal{D}|\theta)$$

$\Rightarrow$

$$-\log P(\mathcal{D}|\theta) =$$

$$-\sum_{i=0}^{n-1} \left[y_i \log p_i + (1-y_i) \log (1-p_i) \right]$$

$\theta_0 + \theta_1 x_i$

$$p_i = \frac{e^{\theta_0 + \theta_1 x_i}}{1 + e^{\theta_0 + \theta_1 x_i}}$$

$$z = \theta_0 + \theta_1 x_1'$$

$$\begin{aligned} \log p_i &= (\theta_0 + \theta_1 x_1') \\ &\quad - \log (1 + e^{\theta_0 + \theta_1 x_1'}) \\ &= z - \log (1 + e^z) \end{aligned}$$

$$\Rightarrow C(\theta) = - \sum_{i=0}^{n-1} [y_i' \cdot z - \log (1 + e^z)]$$

$$\nabla_{\theta} C(\theta) = 0$$

Graphical representation

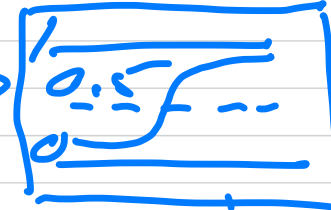
Input
layer

$$z = \theta_0 + \theta_1 x_1$$

x_1

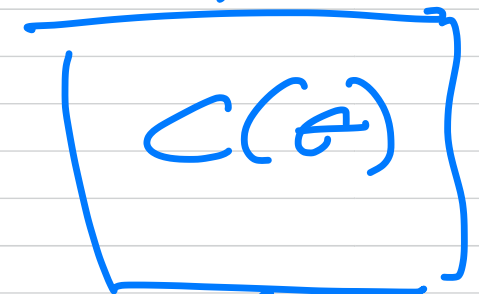
$$\sum(x_1, \theta)$$

$$P(x_1) = p_1$$



Model
Output
layer

$\hat{y}_1$



y_1

simple
perceptron

$$X = \begin{bmatrix} x_{00} & x_{01} & - & - & x_{0p-1} \\ x_{10} & - & - & - & x_{1p-1} \\ \vdots & & & & \\ \boxed{x_{n'0} \quad x_{n'1} \quad - \quad - \quad x_{n'p-1}} \\ \vdots & & & & \\ x_{n-10} & - & - & - & x_{n-1p-1} \end{bmatrix}$$

$$z = \sum_{j=0}^{p-1} x_{n'j} \theta_j$$

(here $z = \theta_0 + \theta_1 x_{n'}$)

$$\frac{\partial C}{\partial \theta_0} = 0 = - \sum_{i=0}^{n-1} (y_i - p_i)$$

$$\frac{\partial C}{\partial \theta_1} = 0 = - \sum_{i=0}^{n-1} x_i (y_i - p_i)$$

$$\vec{\nabla}_{\theta} C(\theta) = -X^T (\vec{y} - \vec{p})$$

$$C(e) = - \sum_{i=0}^{n-1} \underbrace{(y_i' \cdot z - \log(1 + e^z))}_{L_i'}$$

$$\frac{\partial L_i'}{\partial \theta_j'} = \frac{\partial L_i'}{\partial z} \cdot \frac{\partial z}{\partial \theta_j'}$$

$$z = \sum_{j=0}^{P-1} x_{i'j'} \theta_j'$$

in our simpler model

$$z = \theta_0 + \theta_1 x_{i'}$$

$$P_i(x_i) \rightarrow \sigma(z)$$

Sigmoid function

$$\frac{\partial \mathcal{L}_1}{\partial z} = -y_1 + \frac{e^z}{1+e^z} =$$

$$-y_1 + p_1$$

$$\frac{\partial z}{\partial \theta_j} =$$

$$x_{ij}$$

$$\frac{\partial \mathcal{L}_1}{\partial \theta_j} = \frac{\partial \mathcal{L}_1}{\partial z} \frac{\partial z}{\partial \theta_j} = \underline{-x_{ij}(y_1 - p_1)}$$

GD

$$e^{(n+1)} = e^{(n)} - \gamma^{(n)} \nabla_{\theta} C(e^{(n)})$$

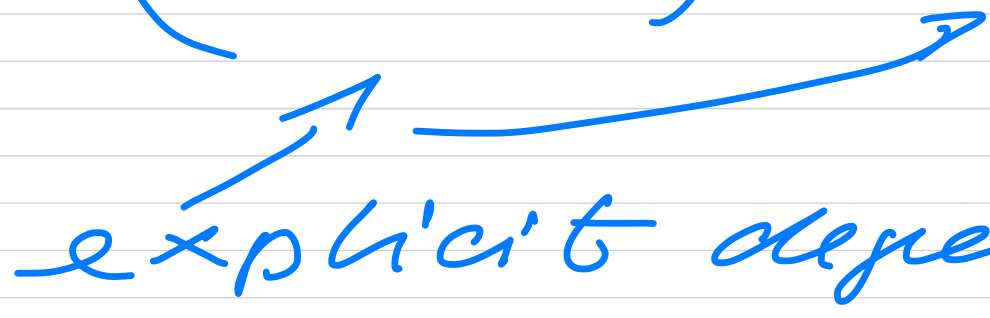
$$\frac{\partial^2 C}{\partial \theta^2} = \underbrace{X^T}_{p \times n} \underbrace{W}_{n \times n} \underbrace{X}_{n \times p} \quad (p \times p)$$

$$W = \begin{bmatrix} w_{00} & & 0 \\ & w_{11} & \\ 0 & & \ddots \\ & & & w_{n-1, n-1} \end{bmatrix}$$

$w_{ii} = p_i(1 - p_i)$

$$P_{x_1} = \frac{e^{\theta_0 + \theta_1 x_1}}{1 + e^{\theta_0 + \theta_1 x_1}}$$

$$\theta^{(n+1)} = \theta^{(n)} - \left(H[\theta^{(n)}] \right)^{-1} \nabla_{\theta} C(\theta^{(n)})$$


 explicit dependence on $\theta^{(n)}$

in linear regression

$$H^{(n)} = \frac{2}{n} X^T X \quad \text{no } \theta\text{-dependence}$$